Medical Hypotheses 142 (2020) 109836

Contents lists available at ScienceDirect

Medical Hypotheses

journal homepage: www.elsevier.com/locate/mehy

Sleep debt induces skeletal muscle injuries in athletes: A promising hypothesis

Luísa de Sousa Nogueira Freitas, Flavia Rodrigues da Silva, Henrique de Araújo Andrade, Renato Carvalho Guerreiro, Fernanda Viegas Paulo, Marco Túlio de Mello, Andressa Silva*

Departamento de Esportes, Universidade Federal de Minas Gerais, Belo Horizonte, Minas Gerais, Brazil

ARTICLE INFO

Keywords:
Sleep
Athletes
Musculoskeletal injuries
Inflammation
Hormones
Protein degradation

ABSTRACT

Sleep is a physiological state and it is fundamental for physical and cognitive recovery of athletes. Due to strenuous training and competitions, athletes may present sleep complaints compromising good quality and quantity of sleep. Studies have related sleep debt to the occurrence of musculoskeletal injuries in athletes, but the mechanisms that can lead to this are not entirely clear. Studies involving animals and humans have shown that poor sleep quality can cause significant changes in hormones and cytokines. Demonstrating that this hormones changes lead to a decrease of testosterone and growth hormone levels and increased cortisol levels, important hormones in the process of protein synthesis and degradation. In athletes, the sport itself is a risk factor of injuries, and sleep debt may result in overtraining syndrome associated with inflammatory markers and ultimately to immune system dysfunction. Thus, we hypothesize that athletes who have sleep debt are more susceptible to musculoskeletal injuries due to increased catabolic pathway signaling, i.e. protein degradation and decreased anabolic pathway signaling, compromising muscle integrity. In this sense, we indicate the relationship between musculoskeletal injuries and sleep debt involving new targets for immunological signaling pathways that start the reduction of the muscle recovery process.

Background

Sleep is a fundamental biological component in the process of energy recovery, cognition and restoration [1–6], as well as metabolic functions and energy balance [7–12], macromolecule biosynthesis [13], removal of toxic substances and metabolic waste [14,15] or prophylactic cellular maintenance [16], and is vital to overall health and well-being [17]. Sleep deprivation (SD) induces damage in both animals and humans, including cognitive [18], immunological, metabolic [19,20] and hormonal [20–23] functions.

Sleep loss results can induce deterioration of many basic functions with physiological changes similar to those seen after stress [24]. In a previous study, our group demonstrated that rats that were submitted to SD presented a loss of body weight [25] and another study corroborates our data indicating that SD activates stress responses in rats [26]. Therefore, it is necessary to detect factors that may be associated with an increased risk of musculoskeletal injuries, such as quality and quantity of sleep, as well as biomarkers.

In this context, considering the peculiarities of each sport and the physical demands of the athlete, the sport itself is considered a risk factor for injury [27]. In addition to the physiological and behavioral impacts already mentioned, a musculoskeletal injury can have significant financial impacts for teams and sponsors, where the average cost of an elite athlete away for a month to recover from an injury is approximately 50,000 euros [28]. Otherwise, it is important to consider that a reduction in sleep quality and quantity may result in an imbalance in the autonomic nervous system function, leading to an increase in the probability of development of overtraining syndrome and inflammatory markers and ultimately to immune system dysfunction [29]. Thus, the hypothesis that characterizes this relationship between musculoskeletal injuries and sleep debt, associated with biomarkers will be raised in this work.

Quality and quantity of sleep and its importance to athletes

To assess sleep quality, it is important to evaluate some variables, such as total sleep time (TST), which reflect the duration of sleep, Sleep Onset Latency (SOL) represents the time taken to fall asleep, the number of awakenings after sleep onset (WASO) indicate sleep fragmentation and sleep efficiency (SE), which is expressed by measuring

* Corresponding author.
E-mail address: andressa@demello.net.br (A. Silva).

https://doi.org/10.1016/j.mehy.2020.109836
Received 17 March 2020; Received in revised form 23 April 2020; Accepted 11 May 2020
0306-9877/ © 2020 Elsevier Ltd. All rights reserved.

the relationship between bedtime and TST [30].

The importance of sleep for athletes have been previously demonstrated by Watson in 2017, who suggested that athletes need a TST between 9 and 10 h to effectively recover from high sports training and physical demands [31]. A recent work evaluated 146 Brazilian Olympic athletes in the preparatory cycle for the Rio 2016 Games. Each athlete underwent a single night polysomnography and answered the UNIFESP Sleep Questionnaire developed and validated by the Sleep Institute/AFIP. This study demonstrated, according to the questionnaire, that 53% had some sleep disorder, where 250 sleep complaints were reported and the most frequent was insufficient sleep/waking up tired. According to the polysomnography men had significantly higher SOL and WASO than women and a lower SE [32]. This scenario illustrates that athletes have poor sleep quality, that could impact recovery and make athletes more susceptible to injury [33].

Sleep duration is known to affect physiological and circadian metabolites and human homeostasis [34]. Regarding athletes, their social and physical environment often causes the lack of restorative sleep [35], where the sleep quality may be altered due to different factors, such as a congested competition calendar, low sleep priority about other training demands, and the small amount of knowledge about the role of sleep in optimizing athletic performance [36,37]. Although sleep is essential to athlete's performance, most of them suffer from sleep insufficiency and due to several competitions, the circadian rhythm and sleep patterns of athletes are often disturbed [31,38,39]. Moreover, intensive training programs result in muscle pain and psychological stress, which makes it more difficult for athletes to have enough sleep [40].

Sleep plays an important role in muscle integrity since during sleep occurs the release of hormones associated with protein synthesis and the reduction of hormones associated with protein degradation [41], maintaining the balance between protein synthesis and degradation. In this context, studies in animals have shown that sleep restriction (SR) or sleep deprivation (SD) results in impairments in this balance [42,43].

Elite athletes generally have a higher prevalence of complaints of insomnia, shorter sleep duration, and poor sleep quality when compared to non-athletes [44]. Studies show that SD or SR can impair acute maximal muscle strength [45]. In a similar way, impair the processes that are responsible for regulation of attention and concentration, which can lead to degraded coordination [46]. Thus, this negative impacts of SD or SR could result in higher musculoskeletal injuries risk. It has shown that Australia's elite soccer athletes, during a season, had low sleep quantity and it was associated with a higher incidence of illness [47]. Athletes can get enough sleep, but with an inadequate sleep quality which represents unrepairable sleep [48]. In these cases, these athletes do not achieve the benefits of restorative sleep and may appear tired and less recovered, not reaching their potential in training and/or competition [49], showing a decline in the performance of sport-specific skills practiced [50].

Paralympic athletes have unique pre-existing medical conditions that predispose them to an increased risk of disease [51]. Besides that, they are also constantly exposed to rigorous training and pressure to stay competitive [52], where increased training loads in combination with the pressure to perform can eventually lead to sports injuries and illnesses that may compromise the health of these athletes [52,53]. A recent study showed that these athletes report a high prevalence of sports-related injuries and illnesses in Paralympic sports [54].

From that perspective, our group conducted a study that assessed the sleep quality, sleepiness, chronotype, and anxiety level of Brazilian Paralympic athletes before the Beijing Paralympic Games in 2008, showing that 83.3% of athletes who had excessive daytime sleepiness, also had poor sleep quality and 72% of athletes who had medium anxiety levels also had poor sleep quality [55]. Subsequently, a group of Chilean researchers determined that sleep quality, insomnia and excessive daytime sleepiness in Chilean Paralympic athletes is a situation that may negatively influence sports performance [56], and that can trigger an increase in musculoskeletal injuries. Thus, considering that sleep quality may interfere with athlete performance and influence muscle recovery [41], the risk of musculoskeletal injuries may become higher [33,49,57].

Hormonal changes due to sleep quality and quantity to the skeletal muscle tissue

Sleep is a physiological state where are observed reduction of responses in the ambient stimulus and occurs rhythmically composed by REM and NON-REM sleep (NREM) [58]. REM is characterized by low amplitude and high-frequency brain activity, similar to the waking state [59]. Also, this stage is characterized by episodes of rapid eye movement and maximum muscle relaxation [59]. During REM sleep occur primarily cognitive restoration, learning, and consolidation of memory, as well as hormonal release spikes such as testosterone [60], which demonstrates the importance of sleep for the consolidation of biological processes in athletes. On the other hand, N-REM sleep, when growth hormone (GH) release occurs [61], is physiologically stable compared to wakefulness and REM sleep [62], being important for tissue reconstitution and physical recovery.

Endogenous central and peripheral circadian oscillators are key organizers of multiple aspects of mammalian physiology, tracking the day-night cycle as well as behavioral and physiologic rhythmicity, and when circadian rhythmicity and sleep homeostasis are misaligned, injuries may occur [63], knowing that sleep is indispensable for the homeostasis of organisms, and is closely associated with circadian rhythms [64] and hormone levels [65].

The rhythmicity circadian process is regulated by endogenous circadian clock located in the suprachiasmatic nucleus (SCN) [66], which in mammals, orchestrates this network for internal time, in which the multiplicity of peripheral clocks exhibit defined but permutable phase relationships to each other and the environmental light, for example, the dark cycle [63]. This system can ensure homeostasis over time; as an example, the hepatic clock's proper timing of compensatory glycogenolysis and glucose export enables maintenance of plasma glucose levels through nighttime fasting [67,68] contributing to muscle protein synthesis. The circadian clock in the SCN acts through action of neurotransmitters and other biomarkers [63], that regulate phase and amplitude of the molecular clock, consequently, it is important for the skeletal muscle homeostasis [63].

Nowadays, exists a lack of established clinical biomarkers to discriminate adequately versus insufficient sleep within an individual, thus, the Sleep Research Society and the National Institutes of Health have recognized an immediate need and potential long-term benefits of identifying the objective, point-of-care biomarkers of insufficient sleep [69,70]. Phospholipids, for example, has been a promising key for the understanding of structural and functional components of cell membranes [71] that involve changes in skeletal muscle phospholipid metabolism and have been linked to altered insulin sensitivity [72,73] leading to new perspectives for research in athletes.

In this sense, a previous work reported that insulin sensitivity was reduced during insufficient sleep [74] and such changes in insulin sensitivity may be linked to some of the altered compounds involved in phospholipid metabolism observed in a recent study in which it was developed biomarker models for sleep loss [75]. Interestingly, that nine out of the 10 gene ontology biological processes associated with the gene-metabolite network derived from the performing biomarker model of this study are linked to either phospholipid metabolism or sphingolipid metabolism, specifically, the sphingolipids ceramide and sphingomyelin were altered during insufficient sleep [75]. In addition to phospholipids, changes in sphingolipids, and notably ceramides have also been linked to inflammation and the risk of metabolic disorders [72,76,77]. These data suggest that the biomarker model may in part be reflecting changes in inflammation and insulin sensitivity associated with insufficient sleep and certainly emerges as a new target for

research in athletes with sleep debt.

During sleep, occurs release of hormones that are associated with protein synthesis [41,78,79]. It is well established that muscle health is strongly influenced by hormonal secretion, where Testosterone, GH and IGF-1 are known to increase the activity of protein synthesis through phosphatidylinositol-3 kinase/protein kinase B pathway (PI3K/Akt). Thus, they are anabolic hormones [80] as mentioned before. On the other hand, corticosterone/cortisol increases ubiquitination and proteasomal degradation pathway [81]. It is important to notice that, previously, our group demonstrated that the skeletal muscle recovery in rats has been compromised by SD, which strongly influences these anabolic and catabolic hormones [82] mentioned above. GH reaches peak plasma concentration 90 min after sleep onset [83], thus, whenever sleep is interrupted, this hormone release decreases [84].

Testosterone levels, however, are related to the first episode of REM sleep at night and have anabolic effects [85], which leads to increased protein synthesis [86]. Thus, a sleep restriction may lead to significant reductions in this hormone with a 10–15% reduction in testosterone levels in young adults in just one week of sleep restriction [87].

Concerning cortisol, its concentration reaches maximum values between 8 and 9 am, having its levels reduced during the day, reaching minimum values at the beginning of sleep [88]. Studies involving changes in sleep patterns have shown that continuous awakenings during nighttime sleep steadily trigger cortisol secretion pulses, increasing their concentrations [89]. In humans, just 1 day (1d) of SD was enough to cause an increase in cortisol levels [90].

It is important to emphasize the rhythmicity of these hormones, where a SR or SD can lead the athlete to sleep during the day, that is, outside the dark cycle of the day. In the daytime period, even if the athlete achieves NREM or REM sleep stages, hormonal release does not occur the same way as occur in the night sleep due it's rhythmic component [91]. Since circadian processes will favor catabolic hormones during the bright cycle of the day, sleep in this period is inadequate for athletes [92–94].

In this context, our group has been investigating the impact of SR and SD on the skeletal muscle system in animal model. The first study hypothesized that SD or SR would lead to an impact on the musculoskeletal system (MSE) of animals through an imbalance between anabolic and catabolic pathways, which would lead to increased protein degradation and decreased protein synthesis [95]. Then our group evaluated the effect of a 96 h REM sleep deprivation on the MSE of rats, finding that after SD there was a reduction in weight and cross-sectional area of the anterior tibial muscle fibers of these animals, as well as an increase in the corticosterone levels and a reduction in serum testosterone levels [42]. Later, our group published a second hypothesis that resistance exercise would be able to minimize or reverse these deleterious effects of SD and/or SR, by activating anabolic pathways that were inhibited by SD and/or SR [41]. In this hypothesis, resistance exercise would activate anabolic pathways, which have their signaling reduced with increased activation of catabolic pathways through SR and/or SD, reducing the imbalance between protein synthesis and degradation [41].

These results led our group to investigate the muscles most affected by SD in rats and it was reported that after a 96 h REM SD that glycolytic muscle fibers are more sensitive to atrophy than oxidative fibers. Furthermore, it was observed that SD, besides leading to an increase in corticosterone and a decrease in testosterone levels, led to a decrease in Insulin Like Growth Hormone Factor 1 (IGF-1) and Tiroxin 4 (T4) [65]. Corroborating the hypothesis that resistance exercise would be able to minimize the deleterious effects of sleep deprivation, our group investigated the alterations underlying SD-induced muscle atrophy and the impact of this condition in rats previously submitted to strength training. SD led to muscle atrophy, but it was attenuated in animals submitted to exercise, obtaining similar results for IGF-1 and corticosterone concentration [65].

Given these findings, our group is trying to investigate the effect of SD in the IGF-1 levels and inflammatory responses, that could be related to recovery from musculoskeletal injury. In 2017, we investigated whether SD may cause a delay in muscle recovery in animals previously submitted to injury and was reported that SD negatively impaired muscle regeneration, where muscle IGF-1 reduction was observed and, in contrast, sleep recovery was able to restore IGF-1 levels, but was insufficient to normalize muscle regeneration [82]. In 2019, we demonstrate that in humans under SD and eccentric exercise-induced muscle injury, there was no delay in muscle recovery, but some inflammatory responses were modified [96].

Musculoskeletal injury in athletes with SD or SR may also involve failures in pro-resolutive mechanisms that commonly occur in inflammation through pro-resolutive mediators: lipoxins [97] e annexin A1 (AnxA1) [98], lipoxin LXA4 [99], resolvins E (RvE) and D (RvD), as well as maresins and protectins, whose functions stand out, among others, by accelerating the resolution of inflammation, reducing the production of pro-inflammatory cytokines and lipid mediators such as prostaglandins and leukotrienes and enhancing the expression of anti-inflammatory mediators [100]. It may involve failures about the deregulation of heat shock proteins (HSPs), a group of cytoprotective proteins that are critical in maintaining protein and cellular homeostasis, that are implicated in adaptations induced by exercise and activated with acute exercise or after chronic physical training regimens [101], which has become beneficial for therapeutic direction as strategies during athlete training [101].

The responses found in animal models have not been completely reproduced in humans, making research in this area important urgent. Given this scenario, our group evaluated the sleep and injury association of 23 high-performance soccer athletes. The sleep behavior was monitored through actigraphy using the Actiwatch 2 (Philips Respironics, Andover, MA), configured with thresholds $>$ 80 counts, for 10 consecutive days. Musculoskeletal injuries record was performed in conjunction with the medical team for six months. Through the actigraphy data our group reported that these athletes generally slept 6.82 h per night. Thus, we found that 44% of the total variance of the total number of injuries and 47% of the severity of injuries can be explained by sleep efficiency [33]. Demonstrating that low-efficiency sleep can influence the musculoskeletal injuries of athletes, but the mechanisms by which this influence occurs are not entirely clear and need to be better elucidated. All these data show an association between sleep and increased incidence of injuries by different mechanisms, highlighting the necessity of more studies involving biomarkers and sleep debt in athletes to avoid musculoskeletal injuries.

Relationship between sleep debt and skeletal muscle injuries in athletes

Injury presents a significant health burden to sporting populations influencing both training and competition performance outcomes [102]. Overtraining syndrome is seen in athletes when they become progressively fatigued and fail to recover from intensive training sessions, and they may suffer from minor infections [103] and chronic or acute sleep loss is directly correlated to athletic injuries, indicating the role of sleep in the regeneration of damaged muscle tissue, as has been showed in previous studies [17,104,105].

Though skeletal muscles have high plasticity and can recover from several types of injuries [105], it is well established that sleep influences physical performance and recovery following strenuous activity [31,106,107]. In addition to the cognitive consequences of sleep loss, impaired physical performance has been reported in athletes, including skill execution, submaximal strength, and muscular power [108]. There is currently a concern to conduct studies involving sleep and muscle injury [109]. Lack of sleep negatively impacts skeletal muscles, and the hormonal imbalance and cytokines may contribute to these effects, especially after the introduction of stimuli that result in greater musculoskeletal damage [96], since high-intensity exercise can affect

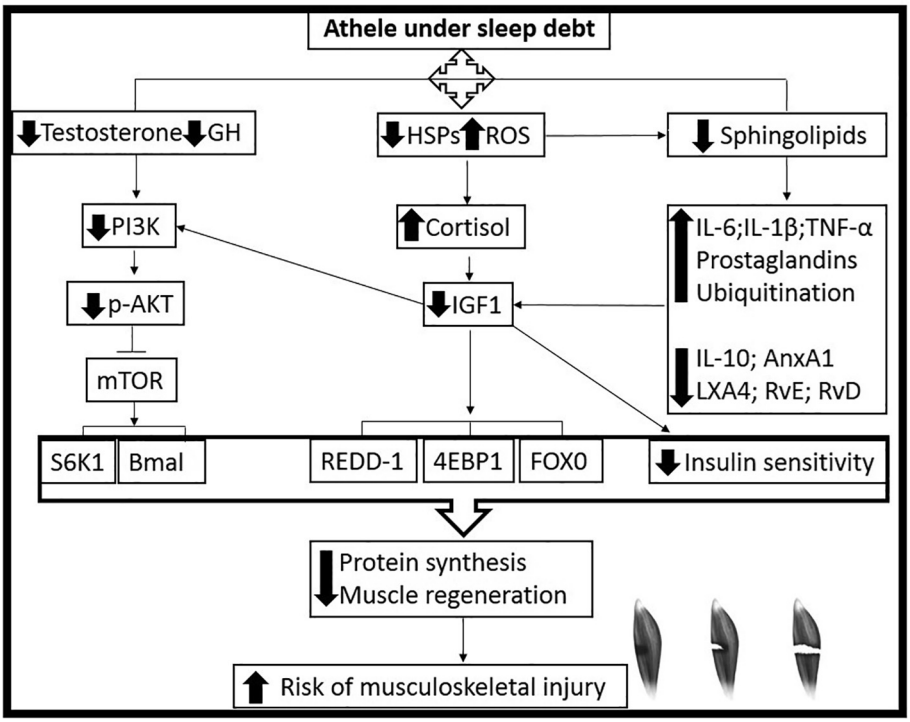

Fig. 1. Mechanism proposed for increased incidence of musculoskeletal injuries in athletes with sleep debt: Athletes could present a reduction of testosterone and GH, which promote a reduction of PI3K activation, and consequently decreased phosphorylation of Akt, which inactivate mTOR signaling, impairing the S6K1 and Bmal activation. Sleep debt in athletes also may decrease HSPs and increase ROS, and increase of cortisol levels, which impair IGF-1 secretion. Consequently, the reduction of IGF-1 increase PI3K inactivation and induces REDD-1, 4EBP1 and FOX0 activation. The sleep debt could impair the sphingolipds morphology on the cell membrane, inducing a disturbance of inflammatory regulation, involving increased levels of: IL-6; IL-1β; TNF-α, activation of prostaglandins, and ubiquitination processes. Besides that, IL-10; AnxA1; LXA4, RvE and RvD levels are signaled to be reduced. The impact over sphingolipds contribute to reduce IGF-1 secretion, reinforcing the pathways previously described and, furthermore, reduce insulin sensitivity. The result of the disturbance on cell signaling is a reduction of protein synthesis and muscle regeneration, and the high risk of musculoskeletal injury. *Legend:* GH: growth hormone, PI3K: phosphatidylinositol-3 kinase, p-Akt: phosphorikated protein kinase B, mTOR: mammalian target of rapamycin, S6K1: Ribosomal Protein S6 Kinase 1, HSPs: heat shock proteins, ROS: reactive oxygen species, IGF-I: insulin-like growth factor I, REDD1: regulated in development and DNA damage responses 1, 4EBP1: eukaryotic translation initiation factor 4-E binding protein 1, FoxO: forkhead transcription factor, IL-6: Interleukin 6, IL-1ß: Interleukin 1 beta, TNFα: tumor necrosis factor alpha, IL-10: Interleukin-10, AnxA1: Annexin A1, LXA4: Lipoxin A4, RvE: resolvin E, RvD: resolvin D.

inflammatory responses, that could cause musculoskeletal injury [17].

From this perspective, when muscle damage occurs, muscle cells release Creatine Kinase (CK) into the blood [17] and a recent study demonstrated that CK is known to be an accurate indicator of muscle damage in clinical practice [110]. Parallel to this, levels of aspartate aminotransferase (AST), alanine aminotransferase (ALT), albumin, lactate dehydrogenase (LDH), besides CK, are markers of muscle damage after strenuous exercise that can be disturbed by sleep loss [111]. Thus, a recent study indicated that the levels of AST and CK activities confirmed muscle impairment in mice with SD and submitted to physical exercise [17]. These findings could support that elite soccer players exhibit lower sleep quality associated with increased number and severity of musculoskeletal injuries [33], as demonstrated by our group recently, showing that the muscle recovery can be compromised by SD, which strongly influences anabolic and catabolic hormones in these athletes [105], as mentioned before.

Recapturing, SD may reduce the secretion of GH, thereby reducing the ability to repair the damaged muscles after strenuous exercise training [17]. These facts previously exposed has been correlated to a muscle clock which develop a crucial role in muscle metabolism, anticipating changes in glucose absorption and oxidation in the transition between sleep and awakening [112]. Thus, a study has shown that BMAL1, a circadian transcription factor located in the skeletal muscle, is capable of stimulating protein gene translation regardless of its transcriptional activity, where the S6 ribosomal protein kinase 1 (S6K1), an important mTOR-dependent translation regulator, rhythmically phosphorylates BMAL1, which in turn promotes protein synthesis [113]. Two animal models of BMAL1 muscle-specific inactivation indicated that the intrinsic muscle clock controls both glucose uptake and glucose metabolism in skeletal muscle. Therefore, the muscle clock plays an important role in preparing skeletal muscle tissue in the transition from rest to the active phase when glucose becomes the predominant fuel for skeletal muscle [112].

The PI3K/Akt pathway mentioned before, which can be stimulated by IGF-1, is one of the important signaling cascades in protein synthesis. It inhibits protein kinase glycogen-3 (GSK-3b) and activates the mammalian target of rapamycin (mTOR) [80]. The mTOR signaling pathway has a wide range of functions, which culminate in hormonal, nutritional, mechanical, hypoxic and cellular stress signals including the regulation of protein synthesis, cell proliferation, apoptosis and autophagy [114]. In animal model studies, SD leads to reduced GH and testosterone production as well as increased corticosterone, [65,115], affecting the protein turnover [41]. Muscle protein turnover is regulated through the activities of the Akt/mechanistic target of rapamycin complex 1 (Akt/mTOR) pathway and the ubiquitin proteasome system (UPS) as well as the lysosomal-autophagic system [41,116,117].

GH secretion and testosterone stimulate muscle growth and recovery through stimulation of anabolic pathways (PI3k-Akt-mTOR) and inhibition of catabolic pathways (REDD1-4EBP1) [65,95]. On the other hand, cortisol plays an important role in protein degradation pathways by activating UPS and inhibiting IGF-1 action and increasing the REDD1 transcription and Forkhead transcription factor (FOXO), resulting in reduced activity of the protein mTOR and its targets [82]. Thus, typical changes in sleep may involve hormonal secretion patterns and may decrease protein synthesis and increase protein degradation, impairing skeletal muscle integrity, leading to injury [82].

Given that, the resolution of inflammation, oxidative stress, and inflammation are inevitable exercise-related processes that can drastically affect athlete's health and performance [118] and it deserves research relevance. About oxidative stress biomarkers, markers of plasma inflammation of 12 professional mixed martial arts (MMA) athletes were measured before and after combat and found that athletes exhibited mild oxidative plasma imbalances after combat, variation in antioxidant capacity in plasma or antioxidant in erythrocytes. However, a clear pro-inflammatory state was detected by the increase in circulating cytokines IL-6, IL-1β and TNF-α and also an abrupt drop in anti-inflammatory cytokine IL-10 [118]. Recently, data published by our group showed a great innovation about this context involving inflammation and SD on skeletal muscle recovery. This study investigated SD after eccentric exercise-induced muscle damage (EEIMD) in healthy,

recreationally active males who did not have any weight-training experience. We demonstrated that SD modifies the profiles of blood hormones and cytokines for IL-1β and TNF-α. However IL-6 increased immediately after the EEIMD protocol when compared to before the protocol and IL-6 showed a tendency to increase in SD. IL-1β was higher at 2 h than before and immediately after the EEIMD protocol in SD, and IL-10 data were not available and no differences were identified between conditions in the free testosterone levels. However, IGF-1, cortisol, and the cortisol/total testosterone ratio were higher in SD than in the group that slept regularly [96]. This reinforces the integration between oxidative stress and inflammation in athletes as well as integration with catabolic and anabolic pathways discussed earlier (detailed in Fig. 1).

Besides that, different perspectives for sleep debt and its impact on skeletal muscle of athletes has been investigated, and recently, it was demonstrated that sleep efficiency was also associated with urea cycle and Krebs cycle metabolite levels and response time during heavy exercise in volleyball athletes, suggesting that sleep quality may also affect amino acid, energy metabolism and cognitive function during heavy exercise [34], directing us to countless possibilities of prevention of skeletal muscle injury.

Thus, we hypothesize that athletes who have poor sleep quality and reduced TST may be more susceptible to musculoskeletal conditions that may trigger injury. Since, SR or non-restorative sleep may impact muscle recovery due to increased catabolic pathway signaling, and decreased anabolic pathway signaling, compromising muscle integrity. For this reason, a new field of research can be considered, seeking to understand in athletes the physiological, biochemical and immunological mechanisms by which lack of sleep can negatively impact in musculoskeletal injuries.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Authors thank the Universidade Federal de Minas Gerais (UFMG); Centro de Treinamento Esportivo (CTE/UFMG); Centro de Estudos em Psicobiologia e Exercício (CEPE); Centro Multidisciplinar em Sonolência e Acidentes (CEMSA); Fundo de Amparo à Pesquisa de Minas Gerais (FAPEMIG); Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq); Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES); Secretaria Especial do Esporte do Ministério da Cidadania (Governo Federal, Brasília, Brazil); Comitê Paralímpico brasileiro (CPB) and Comitê Olímpico do Brasil (COB).

Funding

This study is supported by Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG), Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES).

References

[1] Dinges JDF, Pack F, Williams K, et al. Cumulative sleepiness, mood disturbance, and psychomotor vigilance performance decrements during a week of sleep restricted to 4–5 hours per night. Sleep 1997;20(4):267–77.
[2] Abel T, Havekes R, Saletin JM, Walker MP. Sleep, plasticity and memory from molecules to whole-brain networks. Curr Biol 2013;23(17):774–88.
[3] Tononi G, Cirelli C. Sleep and the price of plasticity: from synaptic and cellular homeostasis to memory consolidation and integration. Neuron 2014;81(3):12–34.
[4] Baum KT, Desai A, Field J, Miller LE, Rausch J, Beebe DW. Sleep restriction worsens mood and emotion regulation in adolescents. J Child Psychol Psychiatry Allied Discip 2014.
[5] Cote KA, Milner CE, Osip SL, Baker ML, Cuthbert BP. Physiological arousal and attention during a week of continuous sleep restriction. Physiol Behav 2008;95(3):353–64.
[6] Gumenyuk V, Roth T, Korzyukov O, Jefferson C, Bowyer S, Drake CL. Habitual short sleep impacts frontal switch mechanism in attention to novelty. Sleep 2011;34(12):1659.
[7] Benington JH, Craig Heller H. Restoration of brain energy metabolism as the function of sleep. Prog Neurobiol 1995;45:347–60.
[8] Berger RJ, Phillips NH. Energy conservation and sleep. Behav Brain Res 1995;69(2):65–73.
[9] Horne J. REM sleep, energy balance and "optimal foraging". Neurosci Biobehav Rev 2009;33:466–74.
[10] Maquet P. Sleep function(s) and cerebral metabolism. Behav Brain Res 1995;69(2):75–83.
[11] Scharf MT, Naidoo N, Zimmerman JE, Pack AI. The energy hypothesis of sleep revisited. Prog Neurobiol 2008;86(3):264–80.
[12] Schmidt MH. The energy allocation function of sleep: A unifying theory of sleep, torpor, and continuous wakefulness. Neurosci Biobehav Rev 2014;47:122–53.
[13] Mackiewicz M, Shockley KR, Romer MA, et al. Macromolecule biosynthesis: a key function of sleep. Physiol Genomics 2007;31(3):441–57.
[14] Varshavsky A. Augmented generation of protein fragments during wakefulness as the molecular cause of sleep: a hypothesis. Protein Sci 2012;21(11):1634–61.
[15] Xie L, Kang H, Xu Q, et al. Sleep drives metabolite clearance from the adult brain. Science 2013;342(6):373–7.
[16] Vyazovskiy VV, Harris KD. Sleep and the single neuron: the role of global slow oscillations in individual cell rest. Nat Rev Neurosci 2013;14:443–51.
[17] Yang DF, Shen YL, Wu C, et al. Sleep deprivation reduces the recovery of muscle injury induced by high-intensity exercise in a mouse model. Life Sci 2019;235:116835.
[18] Goel N, Rao H, Durmer JS, Dinges DF. Neurocognitive consequences of sleep deprivation. Semin Neurol 2009;29:320–39.
[19] Knutson KL, Spiegel K, Penev P, Van Cauter E. The metabolic consequences of sleep deprivation. Sleep Med Rev 2007;11:163–78.
[20] Spiegel K, Leproult R, Van Cauter E. Impact of sleep debt on metabolic and endocrine function. Lancet 1999;354(88):1435–9.
[21] Leproult R, Van Cauter E. Role of sleep and sleep loss in hormonal release and metabolism. Endocrine Develop 2009;17:11–21.
[22] Hipólide DC, Suchecki D, de Carvalho Pinto AP, Chiconelli Faria E, Tufik S, Luz J. Paradoxical sleep deprivation and sleep recovery: effects on the hypothalamic-pituitary-adrenal axis activity, energy balance and body composition of rats. J Neuroendocrinol 2006;18(4):231–8.
[23] Andersen ML, Martins PJF, D'Almeida V, Bignotto M, Tufik S. Endocrinological and catecholaminergic alterations during sleep deprivation and recovery in male rats. J Sleep Res 2005;14(1):83–90.
[24] Bhatnagar S, Vining C, Iyer V, Kinni V. Changes in hypothalamic-pituitary-adrenal function, body temperature, body weight and food intake with repeated social stress exposure in rats. J Neuroendocrinol 2006;18(1):13–24.
[25] Suchecki D, Tufik S. Social stability attenuates the stress in the modified multiple platform method for paradoxical sleep deprivation in the rat. Physiol Behav 2000;68(3):309–16.
[26] Machado RB, Tufik S, Suchecki D. Role of corticosterone on sleep homeostasis induced by REM sleep deprivation in rats. PLoS One 2013;20138(5):1–10.
[27] Gantus MC, Assumpção JD. Epidemiology of the injuries of the locomotor system in basketball athletes. Acta Fisiátrica 2002;9(2):77–84.
[28] Hägglund M, Waldén M, Ekstrand J. Risk factors for lower extremity muscle injury in professional soccer: the UEFA injury study. Am J Sports Med 2013;41(2):327–35.
[29] Fullagar HHK, Duffield R, Skorski S, Coutts AJ, Julian R, Meyer T. Sleep and recovery in team sport: current sleep-related issues facing professional team-sport athletes. Int J Sports Physiol Performance 2015;10:950–7.
[30] Natale V, Plazzi G, Martoni M. Actigraphy in the assessment of insomnia: a quantitative approach. Sleep 2009;32(6):767–71.
[31] Watson AM. Sleep and athletic performance. Curr Sports Med Rep 2017;16(6):413–8.
[32] Silva A, Narciso FV, Rosa JP, et al. Gender differences in sleep patterns and sleep complaints of elite athletes. Sleep Sci 2019:1–7.
[33] Silva A, Narciso FV, Soalheiro I, et al. Poor sleep quality's association with soccer injuries: preliminary data. Int J Sports Physiol Perform 2020:1–6.
[34] Akazawa N, Kobayashi N, Nakamura Y, Kumagai H, Choi Y, Maeda S. Effect of sleep efficiency on salivary metabolite profile and cognitive function during exercise in volleyball athletes. Eur J Appl Physiol 2019;119(10):2215–23.
[35] Kroshus E, Wagner J, Wyrick D, et al. Wake up call for collegiate athlete sleep: narrative review and consensus recommendations from the NCAA interassociation task force on sleep and wellness. British J Sports Med 2019;53:731–6.
[36] Patrick Y, Lee A, Raha O, et al. Effects of sleep deprivation on cognitive and physical performance in university students. SleepBiol Rhythms 2017;15(3):217–25.
[37] Irish LA, Kline CE, Gunn HE, Buysse DJ, Hall MH. The role of sleep hygiene in promoting public health: a review of empirical evidence. Sleep Med Rev 2015;22:23–36.
[38] Rosa JPP, Rodrigues DF, Silva A, et al. Rio olympic games: can the schedule of events compromise athletes' performance? Chronobiol Int 2016;33(4):435–40.
[39] Tuomilehto H, Vuorinen VP, Penttilä E, et al. Sleep of professional athletes: underexploited potential to improve health and performance. J Sports Sci 2017;35(7):704–10.

[40] Hershner SD, Chervin RD. Causes and consequences of sleepiness among college students. Nat Sci Sleep 2014;6:73–84.
[41] Mônico-Neto M, Antunes HKM, Dattilo M, et al. Resistance exercise: a non-pharmacological strategy to minimize or reverse sleep deprivation-induced muscle atrophy. Med Hypotheses 2013;80(6):701—5.
[42] Dattilo M, Antunes HKM, Medeiros A, et al. Paradoxical sleep deprivation induces muscle atrophy. Muscle Nerve 2012;45(3):431–3.
[43] Tufik S, Andersen ML, Bittencourt LRA, de Mello MT. Paradoxical sleep deprivation: Neurochemical, hormonal and behavioral alterations. Evidence from 30 years of research. An Acad Bras Cienc 2009;81(3):521–38.
[44] Leeder J, Glaister M, Pizzoferro K, Dawson J, Pedlar C. Sleep duration and quality in elite athletes measured using wristwatch actigraphy. J Sports Sci 2012;30(6):541–5.
[45] Knowles OE, Drinkwater EJ, Urwin CS, Lamon S, Aisbett B. Inadequate sleep and muscle strength: Implications for resistance training. J Sci Med Sport 2018;21(9):959–68.
[46] Mah CD, Sparks AJ, Samaan MA, Souza RB, Luke A. Sleep restriction impairs maximal jump performance and joint coordination in elite athletes. J Sports Sci 2019;37(17):1981–8.
[47] Fitzgerald D, Beckmans C, Joyce D, Mills K. The influence of sleep and training load on illness in nationally competitive male Australian Football athletes: a cohort study over one season. J Sci Med Sport 2018:8–12.
[48] Vgontzas AN. Understanding Insomnia in the Primary care Setting: a new model. Insomnia Ser 2004;9(2):1–7.
[49] Samuels C. Sleep, recovery, and performance: the new frontier in high-performance athletics. Phys Med Rehabil Clin N Am 2009;20(1):149–59.
[50] Bonnar D, Bartel K, Kakoschke N, Lang C. Sleep interventions designed to improve athletic performance and recovery: a systematic review of current approaches. Sport Med 2018;48(3):683–703.
[51] Janse Van Rensburg DC, Schwellnus M, Derman W, Webborn N. Illness among paralympic athletes: epidemiology, risk markers, and preventative strategies. Phys Med Rehab Clinics North Am 2018;29:185–203.
[52] Soligard T, Schwellnus M, Alonso JM, et al. How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. Br J Sports Med 2016;50(17):1030–41.
[53] McCall A, Davison M, Andersen TE, et al. Injury prevention strategies at the FIFA 2014 World Cup: perceptions and practices of the physicians from the 32 participating national teams. Br J Sports Med. 2015.49(9):p.603–8.
[54] Fagher K, Dahlström Ö, Jacobsson J, Timpka T, Lexell J. Prevalence of sports-related injuries and illnesses in paralympic. Athletes 2019;12(3):271–80.
[55] Silva A, Queiroz SS, Winckler C, et al. Sleep quality evaluation, chronotype, sleepiness and anxiety of Paralympic Brazilian athletes: Beijing 2008 paralympic games. Br J Sports Med 2012.46(2):p.150–4.
[56] Aguero SD, Jofre PA, Standen CV, et al. Calidad del sueno, somnolencia e insomnio en deportistas paralimpicos de elite chilenos. Nutr Hosp 2015;32(6):2832–7.
[57] Fullagar HHK, Duffield R, Skorski S, et al. Sleep, travel, and recovery responses of national footballers during and after long-haul international air travel. Int J Sports Physiol Perform. 2016 11(1):p.86–95.
[58] Dement W, Kleitman N. Cyclic variations in EEG during sleep and their relation to eye movements, body motility, and dreaming. Electroencephalogr Clin Neurophysiol 1957;9(4):673–90.
[59] Antle MC. Sleep: neural systems. Int Ency Social Behavioral Sci 2015;21(2):87–93.
[60] Hagenauer MH, Ku JH, Lee TM. Chronotype changes during puberty depend on gonadal hormones in the slow-developing rodent, Octodon degus Megan. Horm Behav 2012;1(3):233–45.
[61] Lastella M, Roach GD, Halson SL, Martin DT, West NP, Sargent C. The impact of a simulated grand tour on sleep, mood, and well-being of competitive cyclists. J Sports Med Phys Fitness 2015;55(12):1555–64.
[62] Martins PJF, De Mello MT, Tufik S. Exercício e sono. Rev Bras Med Esporte 2001;7(1):28–36.
[63] Schwartz WJ, Klerman EB. Circadian neurobiology and the physiologic regulation of sleep and wakefulness. Neurol Clin 2019;37:475–86.
[64] Goel N, Basner M, Rao H, Dinges DF. Circadian rhythms, sleep deprivation, and human performance. Prog Mol Biol Transl Sci 2013;119:155–90.
[65] Monico-Neto M, Antunes HKM, Lee KS, et al. Resistance training minimizes catabolic effects induced by sleep deprivation in rats. Appl Physiol Nutr Metab 2015;40(11):1143–50.
[66] Durmer JS, Dinges DF. Neurocognitive consequences of sleep deprivation. Semin Neurol 2005;25:117–29.
[67] Lamia KA, Storch KF, Weitz CJ. Physiological significance of a peripheral tissue circadian clock. Proc Natl Acad Sci USA 2008;105(39):15172–7.
[68] Qian J, Scheer FAJL. Circadian system and glucose metabolism: implications for physiology and disease. Trends Endocrinol Metab 2016;27:282–93.
[69] Mullington JM, Abbott SM, Carroll JE, et al. Developing biomarker arrays predicting sleep and circadian-coupled risks to health. Sleep 2016;39(4):727–36.
[70] Depner CM, Cheng PC, Devine JK, et al. Wearable technologies for developing sleep and circadian biomarkers: a summary of workshop discussions. Sleep 2020;43(2):1–13.
[71] Matthews BW. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. BBA – Protein Struct 1975;405(2):442–51.
[72] Pérez-Matos MC, Morales-álvarez MC, Toloza FJK, et al. The phospholipid linoleoylglycerophosphocholine as a biomarker of directly measured insulin resistance. DiabetesMetab J 2017;41(6):466–73.
[73] Funai K, Lodhi IJ, Spears LD, et al. Skeletal muscle phospholipid metabolism regulates insulin sensitivity and contractile function. Diabetes 2016;65(2):358–70.
[74] Eckel RH, Depner CM, Perreault L, et al. Morning circadian misalignment during short sleep duration impacts insulin sensitivity. Curr Biol. 2015. 25(22):p. 3004–10.
[75] Depner CM, Cogswell DT, Bisesi PJ, et al. Developing preliminary blood metabolomics-based biomarkers of insufficient sleep in humans. Sleep 2020:1–38.
[76] Turpin-Nolan SM, Hammerschmidt P, Chen W, et al. CerS1-derived C18:0 ceramide in skeletal muscle promotes obesity-induced insulin resistance. Cell Rep 2019;26(1):1–10.
[77] Chaurasia B, Tippetts TS, Monibas RM, et al. Targeting a ceramide double bond improves insulin resistance and hepatic steatosis. Science 2019;365(6451):386–92.
[78] Van Cauter E, Holmbäck U, Knutson K, et al. Impact of sleep and sleep loss on neuroendocrine and metabolic function. Horm Res 2007;67(1):1–9.
[79] Davenne D. Sleep of athletes – problems and possible solutions. Biol Rhythm Res 2009;40(1):45–52.
[80] Bodine SC, Stitt TN, Gonzalez M, et al. Akt/mTOR pathway is a crucial regulator of skeletal muscle hypertrophy and can prevent muscle atrophy in vivo. Nat Cell Biol 2001;3(11):1014–9.
[81] Bodine SC, Latres E, Baumhueter S, et al. Identification of ubiquitin ligases required for skeletal Muscle Atrophy. Science 2001;94(47):1704–8.
[82] Mônico-Neto M, Dáttilo M, Ribeiro DA, et al. REM sleep deprivation impairs muscle regeneration in rats. Growth Factors 2017.35(1):p.12–8.
[83] Takahashi Y, Kirnis DM, Daughaday WH. Growth hornone secretion during sleep. J Clin Investig 1968;47(9):2079–90.
[84] Van Cauter E, Copinschi G. Interrelationships between growth hormone and sleep. Growth Horm IGF Res 2000;10(1):57–62.
[85] Crewther B, Keogh J, Cronin J, Cook C. Possible stimuli for strength and power adaptation: acute hormonal responses. Sports Med 2006;36:215–38.
[86] Wittert G. The relationship between sleep disorders and testosterone. Endocrinol Diab Obesity 2014;21:239–43.
[87] Leproult R, Van Cauter E. Effect of 1 week of sleep restriction on testosterone levels in young healthy men. JAMA 2011;30(5):2173–4.
[88] Chan S, Debono M. Review: Replication of cortisol circadian rhythm: New advances in hydrocortisone replacement therapy. Ther Adv Endocrinol Metab. 2010. 1(3):p.129–38.
[89] Swinbourne R, Miller J, Smart D, Dulson DK, Gill N. The effects of sleep extension on sleep, performance, immunity and physical stress in rugby players. Sports 2018;6(42):1–10.
[90] Leproult R, Copinschi G, Buxton O, Van Cauter E. Sleep Loss Results in an Elevation of Cortisol Levels the Next Evening. Sleep. 1997. 20(10):p.865–70.
[91] Kim TW, Jeong J-H, Hong S-C. The impact of sleep and circadian disturbance on hormones and metabolism. Int J Endocrinol 2015;2015:1–9.
[92] Facer-Childs E, Brandstaetter R. The impact of circadian phenotype and time since awakening on diurnal performance in athletes. Curr Biol 2015;25(4):518–22.
[93] Mougin F, Bourdin H, Simon-Rigaud ML, Nguyen Nhu U, Kantelip JP, Davenne D. Hormonal responses to exercise after partial sleep deprivation and after a hypnotic drug-induced sleep. J Sports Sci 2001;19(2):89–97.
[94] Donald CM, Moore J, McIntyre A, Carmody K, Donne B. Acute effects of 24-h sleep deprivation on salivary cortisol and testosterone concentrations and testosterone to cortisol ratio following supplementation with caffeine or placebo. Int J Exerc Sci 2017;10(11):108–20.
[95] Dattilo M, Antunes HKM, Medeiros A, et al. Sleep and muscle recovery: endocrinological and molecular basis for a new and promising hypothesis. Med Hypotheses 2011;77(2):210–22.
[96] Dáttilo M, Antunes HKM, Galbes NMN, et al. Effects of sleep deprivation on acute skeletal muscle recovery after exercise. Med Sci Sports Exerc 2020;52(2):507–14.
[97] Serhan CN, Hamberg M, Samuelsson B. Lipoxins: Novel series of biologically active compounds formed from arachidonic acid in human leukocytes. Proc Natl Acad Sci USA 1984;81(17):5335–9.
[98] D'Acquisto F, Piras G, Rattazzi L. Pro-inflammatory and pathogenic properties of Annexin-A1: the whole is greater than the sum of its parts. Biochem Pharmacol 2013;85(1):1213–8.
[99] Romano M, Cianci E, Simiele F, Recchiuti A. Lipoxins and aspirin-triggered lipoxins in resolution of inflammation. Eur J Pharmacol 2015;760:49–63.
[100] Serhan CN. Treating inflammation and infection in the 21st century: New hints from decoding resolution mediators and mechanisms. FASEB J 2017;31:1273–88.
[101] Henstridge DC, Febbraio MA, Hargreaves M. Heat shock proteins and exercise adaptations. Our knowledge thus far and the road still ahead. J Appl Physiol Am Physiol Soc 2016;120:683–91.
[102] Hulme A, Nielsen RO, Timpka T, Verhagen E, Finch C. Risk and protective factors for middle- and long-distance running-related injury. Sports Med 2017;47:869–86.
[103] Budgett R. Fatigue and underperformance in athletes: the overtraining syndrome. Br J Sports Med 1998;32(2):107–10.
[104] Chennaoui M, Arnal PJ, Sauvet F, Léger D. Sleep and exercise: a reciprocal issue? Sleep Med Rev 2015;20:59–72.
[105] Dattilo M, Antunes HKM, Medeiros A, et al. Sleep and muscle recovery: endocrinological and molecular basis for a new and promising hypothesis. Med Hypotheses 2011;77(2):220–2.
[106] Watson A, Brickson S. Impaired sleep mediates the negative effects of training load on subjective well-being in female youth athletes. 2018. 10(3): p.244–9.
[107] Milewski MD, Skaggs DL, Bishop GA, et al. Chronic lack of sleep is associated with increased sports injuries in adolescent athletes. J Pediatr Orthop 2014;34(2):129–33.
[108] Fullagar HHK, Skorski S, Duffield R, Hammes D, Coutts AJ, Meyer T. Sleep and athletic performance: the effects of sleep loss on exercise performance, and physiological and cognitive responses to exercise. Sports Med 2014;45(2):161–86.
[109] Burke TM, Lisman PJ, Maguire K, et al. Examination of sleep and injury among

college football athletes. J Strength Cond Res 2019:1–8.
[110] Hsiao CY, Hsu YJ, Tung YT, Lee MC, Huang CC, Hsieh CC. Effects of Antrodia camphorata and panax ginseng supplementation on anti-fatigue properties in mice. J Vet Med Sci 2018;80(2):284–91.
[111] Nathwani RA, Pais S, Reynolds TB, Kaplowitz N. Serum alanine aminotransferase in skeletal muscle diseases. Hepatology 2005;41:380–2.
[112] Schiaffino S, Blaauw B, Dyar KA. The functional significance of the skeletal muscle clock : lessons from Bmal1 knockout models CRY. Skelet Muscle 2016;6(3):1–9.
[113] Lipton JO, Yuan ED, Boyle LM, et al. The circadian protein BMAL1 regulates translation in response to S6K1-mediated phosphorylation. Cell 2015;161(5):1138–51.
[114] Hornberger TA, Sukhija KB, Chien S. Regulation of mTOR by mechanically induced signaling events in skeletal muscle. Cell Cycle 2006;5(1):1391–6.
[115] Giampá SQC, Monico-Neto M, De Mello MT, et al. Paradoxical sleep deprivation causes cardiac dysfunction and the impairment is attenuated by resistance training. PLoS ONE 2016;11(11):1–22.
[116] Dodd KM, Tee AR. Leucine and mTORC1: a complex relationship. Am J Physiol – Endocrinol Metab 2012;30(2):1329–42.
[117] Schiaffino S, Dyar KA, Ciciliot S, Blaauw B, Sandri M. Mechanisms regulating skeletal muscle growth and atrophy. FEBS J 2013;280:4294–314.
[118] Gomes-Santos JAF, Lambertucci RH, Vardaris CV, et al. Early signs of inflammation with mild oxidative stress in mixed martial arts athletes after simulated combat. J Strength Cond Res 2019:1–7.